The Perks of Being a First Mover

Does White's Advantage in Chess Change with Rating, Opening and Time Control?

CS215 Open Project · Final Report · Group 02

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https://github.com/cs215-cse-iitb/dai-aug-2026-openproject-group-2

Section 1: Meta Information

1. Open Project Title	The Perks of Being a First Mover: Does White's Advantage in Chess Change with Rating, Opening and Time Control?
2. Group ID	"02"
3. Mentor Name	Namish Bansal
4. Group Members and Roll Numbers	Sanidhya (25b0944); Daksh Maahor (25b0974); Sahil (25b0991); VadaPavJai (25b1030)
5. Honor Code and Discussions	We did not consult or work with anyone outside our group. All consultations were from LLMs as noted below.
6. LLM Usage Links	• https://claude.ai/share/041dbe60-e304-4ada-bc7e-8d070f0b5a94 • https://claude.ai/share/1bb9222d-6e23-4b2d-8803-6cdff0f0e31c • https://claude.ai/share/49f8001a-bc92-478f-8a8b-d5a4662d43a4
7. Video Presentation	https://drive.google.com/file/d/1d1iBmvHLGC9XuQnEBNOylZaHz9B8BAjR/view
Slides	https://docs.google.com/presentation/d/10hRm3HRv37sTbiAkR7k4MXJolaaZo3Bj/edit?usp=sharing
Tracking Sheet	https://docs.google.com/spreadsheets/d/1pvTllHR4a1Sl7GHtjKnvLeb-zmmHaM3BjltyaDPX9-Q
8. GitHub Repository	https://github.com/cs215-cse-iitb/dai-aug-2026-openproject-group-2
9. GitHub Handles and Initial Commit	
Sanidhya	https://github.com/SS3642 October 21, 2026
Daksh Maahor	https://github.com/Daksh-Maahor November 3, 2026
Sahil	https://github.com/Sahil-M-Deo October 17, 2026
	https://github.com/Sahil-Deo-IITB November 3, 2026
VadaPavJai	https://github.com/nauseated-potato October 18, 2026
10. Contribution Breakdown	Unequal (25b0944: 100%, 25b0974: 100%, 25b0991: 100%, 25b1030: 100%, LLM: 25%)

Work split (for reference)

Sanidhya	Statistical modelling: fitting h, confidence intervals and the z-tests.
Daksh Maahor	Data cleaning and opening classification; win/draw/loss rates.
Sahil	Data engineering: the C++ streaming parser and the Colab pipeline.
VadaPavJai	Visualisation: figures, the Plotly dashboard and the report.
Claude	Progress spreadsheet creation, safety checks in the Colab pipeline code, and final report checks and edits.

Section 2: Technical Content

Core Problem Statement

We measure the first-move advantage on Lichess. Instead of raw win rates, we fit White's advantage as a number of *rating points*, h, after correcting each game for the rating gap between the players. Under the Elo model a player d points above the opponent scores $E(d) = 1/(1+10^{-d/400})$; we give White an extra h points, $E_i(h) = 1/(1+10^{-(R_i^{\mathrm{W}}+h-R_i^{\mathrm{B}})/400})$, and choose the h that makes expected scores equal actual scores ($S_i = 1, \frac{1}{2}, 0$). This is the maximum-likelihood estimate (Newton's method; standard error from the Fisher information). Overall, $h = +11.9 \pm 0.1$ rating points. We test three hypotheses:

- **H1 (Rating).** h peaks at intermediate ratings and shrinks at 2200+.
- **H2 (Openings).** Four aggressive 1.e4 gambits (King's, Evans, Danish, Smith-Morra) carry a larger h than four drawish openings (Petrov, Berlin, QGD, French Exchange), and the drawish ones have more draws at every rating.
- **H3 (Time control).** h grows from Bullet to Classical.

Data Source(s)

Lichess Open Database (`https://database.lichess.org`, CC0), standard rated games for 2024-01 and 2024-02: one zstd-compressed PGN per month, tens of GB each, openly downloadable. We use seven of 17 header tags: `Result`, `WhiteElo`, `BlackElo`, `Event`, `TimeControl`, `Termination`, `Opening`. We read 190,562,735 games and keep 162,175,412 (85.1%).

Month	Games read	Games kept	Kept
2024-01	98,994,760	84,177,790	85.0%
2024-02	91,567,975	77,997,622	85.2%
Total	190,562,735	162,175,412	85.1%

Tools and Software Packages

A C++17 program streams the Lichess zstd files, reads only the header lines, and counts games by opening, speed, 100-point rating bracket and rating gap (two months, two threads, run on Google Colab). It writes `counts.csv` (a few MB); all tables, tests, figures, the report (LaTeX with pgfplots) and the interactive Plotly dashboard (`output/final/dashboard.html`) are rebuilt from it in seconds. The pipeline is compiled with `g++ -O2 -std=c++17 -pthread`. In Python, `pandas` and `NumPy` refit every h from `counts.csv` independently of the C++ code and match all of `output.csv` and `openings.csv` (`check_fit.py`), and add two extra summaries (`h_as_score.py`, `rating_mix.py`); the Colab download uses `requests`. The dashboard uses `plotly.js`. The Colab workflow installs the system library `libzstd-dev`, and the PDF build installs the LaTeX packages `texlive-latex-recommended`, `texlive-latex-extra`, `texlive-fonts-recommended` and `texlive-pictures`.

India Focus

Chess originated in India as *chaturanga*, and India today has one of the world's largest and fastest-growing chess communities, much of it playing online on platforms like Lichess. Lichess records carry no nationality, so our sample is global, and the results describe the online rating pools Indian players compete in.

Challenges

Cleaning and design. We drop games with no result (6,007), abandoned games (502,851), missing ratings (0) and rating gaps above 100 (27,878,465). Games go into 100-point brackets by average rating and into a speed from `Event`. Openings are matched by Lichess name prefix (excluding gambit side-lines inside the drawish

openings) and cover 11,367,920 games (7.0%). With millions of games, even small effects are statistically tiny, so we report effect sizes with 95% CIs. The main technical hurdle was the data volume: monthly Lichess PGN files are tens of GB, so we streamed them directly through a C++ parser with zstd decompression and per-file threading. A second hurdle was naming inconsistencies across opening side-lines and gambit variants, which made a fixed-prefix rule necessary.

Results

Overall

Over all kept games White wins 49.7%, draws 4.0% and loses 46.3%. White's average score is 0.517 against 0.500 predicted by the ratings alone, which is $h = +11.9 \pm 0.1$ rating points.

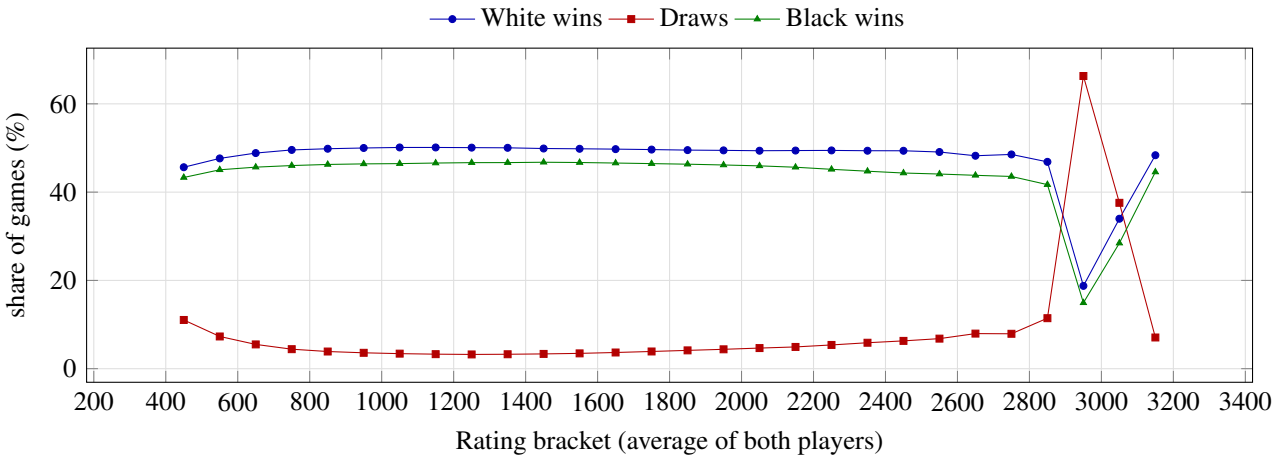


Figure 1: Share of games won by White, drawn and won by Black, by rating (all speeds).

Does the advantage peak at intermediate ratings? (H1)

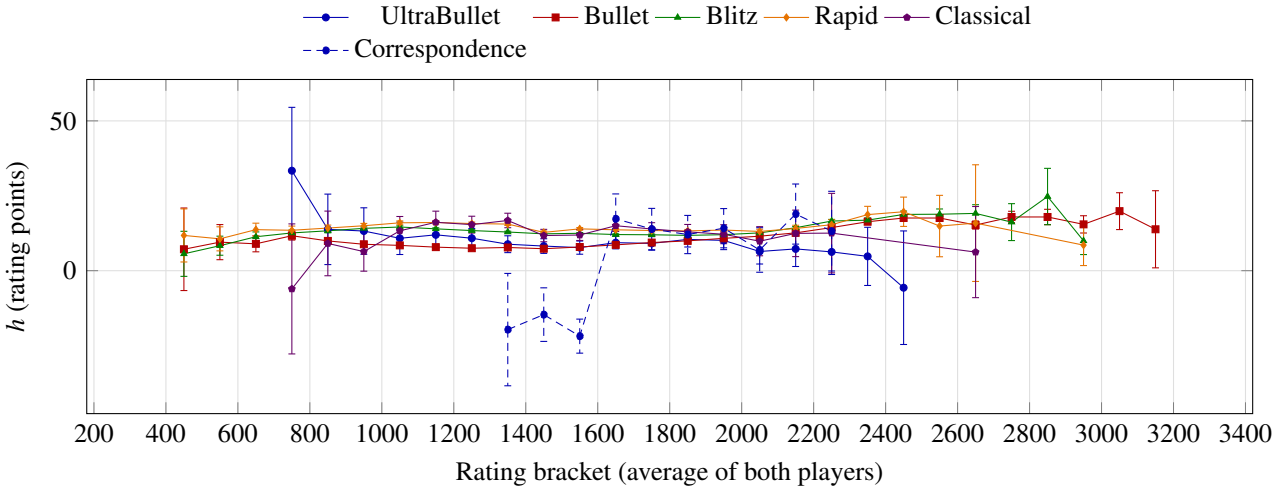


Figure 2: h in each 100-point rating bracket, by speed, with 95% CIs. Brackets with fewer than 1,000 games are left out.

H1 is not supported: h does not have the expected inverted-U shape. It rises from roughly 8 points at 450 to about 18 points at 2400–2600, with no clear peak at intermediate ratings.

Openings (H2)

Over all speeds, h for the aggressive openings minus h for the drawish openings is $+0.2 \pm 0.4$ rating points ($z = 1.0$, $p = 0.298$).

Table 1: White's advantage by opening (all speeds). Avg rating = average rating of the games.

Opening	Games	Avg rating	White	Draw	Black	h (pts)
King's Gambit	2,282,379	1633	51.3%	3.0%	45.6%	$+18.4 \pm 0.5$
Evans Gambit	165,739	1680	51.9%	3.3%	44.8%	$+24.0 \pm 1.7$
Danish Gambit	484,805	1635	53.8%	3.2%	43.0%	$+37.1 \pm 1.0$
Smith-Morra Gambit	1,136,895	1727	49.6%	3.6%	46.8%	$+11.2 \pm 0.6$
Petrov's Defense	2,071,985	1460	50.4%	3.9%	45.7%	$+15.5 \pm 0.5$
Berlin Defense	579,944	1544	50.3%	5.6%	44.1%	$+20.8 \pm 0.9$
Queen's Gambit Declined	3,660,887	1729	51.3%	4.5%	44.2%	$+23.9 \pm 0.4$
French Exchange	985,286	1765	48.1%	4.9%	47.1%	$+4.2 \pm 0.7$
Aggressive (all 4)	4,069,818	1662	51.2%	3.2%	45.6%	$+18.8 \pm 0.3$
Drawish (all 4)	7,298,102	1643	50.5%	4.4%	45.0%	$+18.6 \pm 0.3$
Other openings	141,698,824	1642	49.5%	4.0%	46.5%	$+10.7 \pm 0.1$

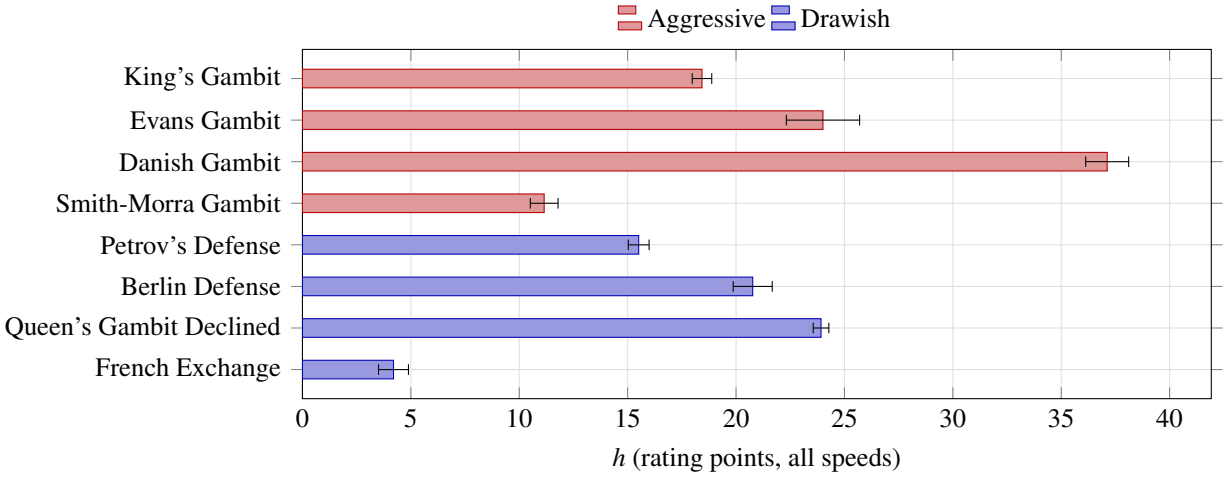


Figure 3: h for each opening (all speeds) with 95% CIs.

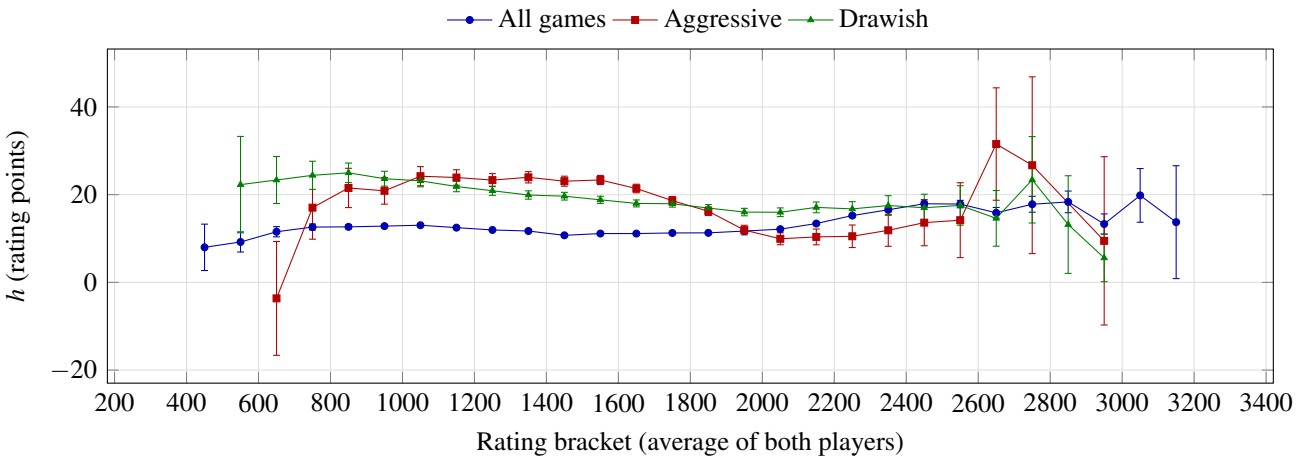


Figure 4: h by rating for the aggressive and the drawish openings, with 95% CIs.

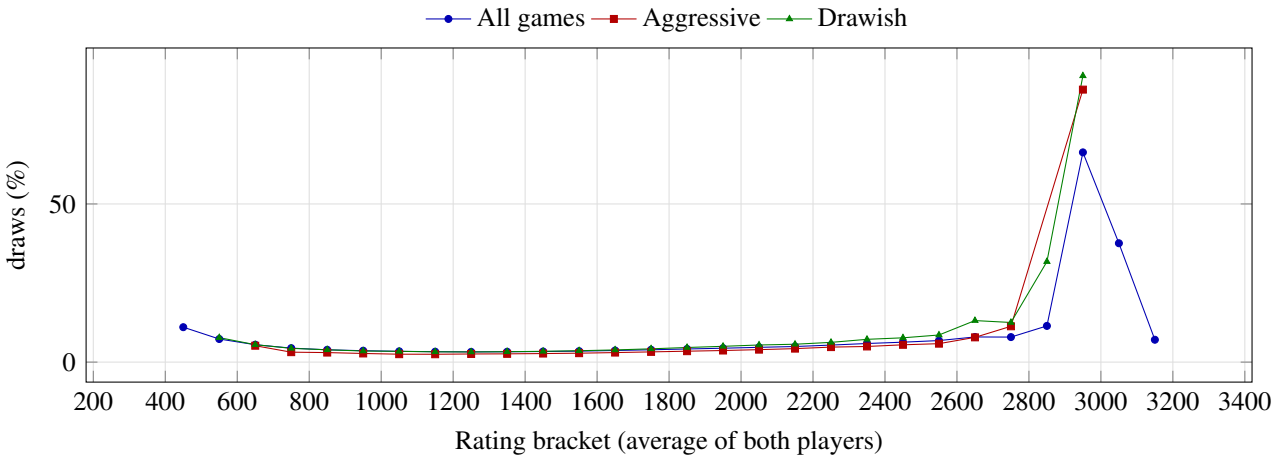


Figure 5: Draw rate by rating for the aggressive and the drawish openings.

Table 2: Aggressive vs drawish openings: h for each group, and the difference with its z-test (rating points, 95% CI), for all speeds and for each of the main speeds.

Speed	Aggressive	Drawish	Difference	z	p
All speeds	$+18.8 \pm 0.3$	$+18.6 \pm 0.3$	$+0.2 \pm 0.4$	+1.0	0.298
Bullet	$+16.9 \pm 0.6$	$+16.3 \pm 0.4$	$+0.6 \pm 0.8$	+1.7	0.097
Blitz	$+19.5 \pm 0.5$	$+18.9 \pm 0.4$	$+0.6 \pm 0.6$	+2.0	0.044
Rapid	$+20.3 \pm 0.9$	$+22.4 \pm 0.6$	-2.1 ± 1.1	-3.7	1.8×10^{-4}

H2 is not supported, although the drawish openings do have a higher draw rate than the aggressive ones (4.4% vs 3.2% overall), which matches the idea that drawish openings are more solid. But the opening-type effect on White's advantage is weak: aggressive openings are only $+0.2 \pm 0.4$ rating points higher than drawish openings, and the difference is not meaningful ($z = 1.0$, $p = 0.298$). So the data do not show a large or consistent White advantage from playing aggressive openings once ratings are accounted for.

Time controls (H3)

Table 3: White's advantage by speed (all openings). White / Draw / Black = share of games won by White, drawn, won by Black. h with 95% CI.

Speed	Games	Avg rating	White	Draw	Black	h (pts)
All speeds	162,175,412	1638	49.7%	4.0%	46.3%	$+11.9 \pm 0.1$
UltraBullet	722,976	1611	50.3%	2.0%	47.8%	$+9.1 \pm 0.8$
Bullet	59,945,392	1716	49.9%	3.0%	47.1%	$+9.9 \pm 0.1$
Blitz	77,918,475	1613	49.6%	4.4%	46.0%	$+12.8 \pm 0.1$
Rapid	22,650,621	1520	49.6%	4.9%	45.5%	$+14.2 \pm 0.1$
Classical	857,174	1569	49.1%	5.7%	45.2%	$+13.4 \pm 0.7$
Correspondence	80,774	1789	47.5%	5.7%	46.8%	$+2.8 \pm 2.4$

H3 is only partly supported. The advantage does increase with slower time controls, from +9.9 in Bullet to +12.8 in Blitz and +14.2 in Rapid, but levels off in Classical ($+13.4 \pm 0.7$, not meaningfully different from Rapid) and is lower in Correspondence ($+2.8 \pm 2.4$, only 80,774 games). The effect is small: about 4 points from Bullet to Rapid.

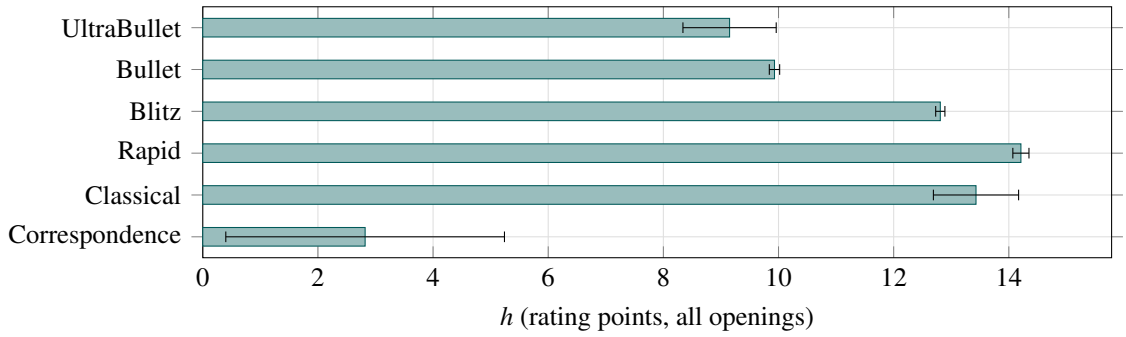


Figure 6: h for each speed (all openings) with 95% CIs.

Table 4: Each speed against the next faster one: h for both, and the difference (slower minus faster) with its z-test (rating points, 95% CI).

Speeds	Slower	Faster	Difference	z	p
Bullet vs UltraBullet	$+9.9 \pm 0.1$	$+9.1 \pm 0.8$	$+0.8 \pm 0.8$	+1.9	0.060
Blitz vs Bullet	$+12.8 \pm 0.1$	$+9.9 \pm 0.1$	$+2.9 \pm 0.1$	+47.8	$< 10^{-300}$
Rapid vs Blitz	$+14.2 \pm 0.1$	$+12.8 \pm 0.1$	$+1.4 \pm 0.2$	+16.9	9.6×10^{-64}
Classical vs Rapid	$+13.4 \pm 0.7$	$+14.2 \pm 0.1$	-0.8 ± 0.8	-2.0	0.042
Correspondence vs Classical	$+2.8 \pm 2.4$	$+13.4 \pm 0.7$	-10.6 ± 2.5	-8.2	2.3×10^{-16}

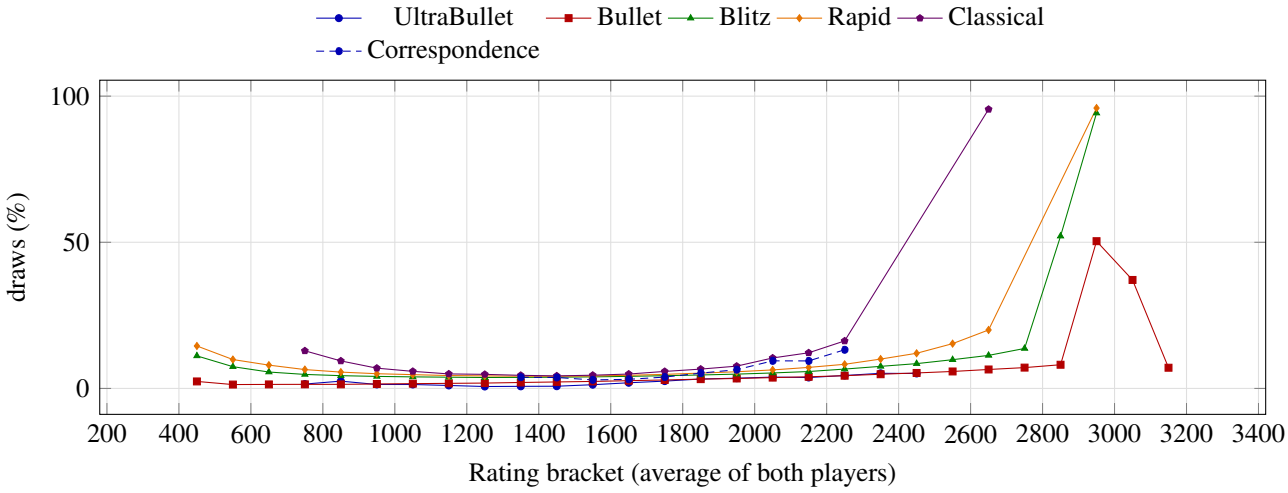


Figure 7: Draw rate by rating for each speed.

Limitations

- *Choosing an opening is not random.* Players who play the King's Gambit or the Berlin differ in style and preparation, and the opening is chosen by one player (White for the gambits and the French Exchange, Black for the Petrov, Berlin and QGD). Our h corrects for the rating gap in each game, but the results show association, not that the opening causes the difference.
- *The groups have different players.* The comparison of the two opening groups doesn't adjust for their players' ratings. The "Avg rating" column and Figure 4 (which compares the groups bracket by bracket) show how much that matters.
- *Ratings are noisy*, especially for new accounts, which blurs the correction for the rating gap. Lichess ratings also differ from FIDE ratings, so "2200 on Lichess" is not a master.
- *Games are not independent.* Active players contribute thousands of games, so the real uncertainty is somewhat larger than our confidence intervals.

Conclusion

Over 162,175,412 Lichess games from 2024-01 and 2024-02, we measured White's first-move advantage as h, the number of rating points it is worth. Our findings:

- **Overall.** White's first move is worth $h = +11.9 \pm 0.1$ rating points. White scores 0.517 where the ratings alone predict 0.500, so the advantage is small but consistent.
- **Rating (H1): not supported.** h does not peak at intermediate ratings and shrink at the top. It rises from about 8 points at 450 to 13 at 1050, dips to about 11 around 1450–1850, then climbs to about 18 at 2400–2600. The draw rate also rises with rating (3.3% to 6.3%), so draws do not explain a decline that is not there.
- **Openings (H2): not supported.** The drawish openings do have more draws than the aggressive ones in every rating bracket (4.4% vs 3.2% overall). But White's edge is no bigger: the aggressive gambits have an h that is $+0.2 \pm 0.4$ points higher than the drawish openings ($z = 1.0$, $p = 0.298$), i.e. no real difference. Within each group the openings vary far more (Danish Gambit +37.1, French Exchange +4.2).
- **Time control (H3): partly supported.** h rises from +9.9 in Bullet to +12.8 in Blitz and +14.2 in Rapid, but levels off in Classical ($+13.4 \pm 0.7$, not meaningfully different from Rapid) and is lower in Correspondence ($+2.8 \pm 2.4$, only 80,774 games). The effect is small: about 4 points from Bullet to Rapid.
- **Caveat.** These are associations, not causal effects. Players choose their openings, and the two opening groups differ in rating (see Limitations).

Section 3: Frequently Answered Questions (FAQ)

Q1. In practical terms, how big is White's edge? *(objective)*
$h = +11.9 \pm 0.1$ means White scores about 51.7% against an equally rated opponent, 1.71 percentage points above 50%. It ranges from $+1.43$ points in Bullet to $+2.04$ in Rapid.

Q2. Which openings give White the largest and smallest edge? *(objective)*
Largest: Danish Gambit, $h = +37.1 \pm 1.0$ (White scores 5.3 points above 50%). Smallest: French Exchange, $+4.2 \pm 0.7$ (0.6 points). The spread inside each group is far larger than the $+0.2$ gap between the aggressive and drawish groups.

Q3. Are aggressive and drawish openings really equal at every rating? *(objective)*
No, they only cancel out overall. Aggressive openings are ahead by $+4.5 \pm 1.3$ points at 1500–1600, while drawish ones are ahead by 6.0 ± 1.7 at 2000–2100 and 6.7 ± 2.2 at 2100–2200.

Q4. How are the $\pm$ intervals for h computed? *(methodological)*
With $E_i(h) = 1/(1+10^{-(d_i+h)/400})$ and $S_i \in \{1, \frac{1}{2}, 0\}$, the log-likelihood is $\ell(h) = \sum_i [S_i \ln E_i + (1-S_i)\ln(1-E_i)]$ and $\ell''(h) = -K^2 \sum_i E_i(1-E_i)$, with $K = \ln 10/400$. The Fisher information is $I(\hat{h}) = -\ell''(\hat{h})$; for large samples $\hat{h} \approx \mathcal{N}(h, 1/I(\hat{h}))$, so SE $= 1/\sqrt{I(\hat{h})}$ and the 95% CI is $\hat{h} \pm 1.96\,\text{SE}$ (overall SE $= 0.028$, i.e. ± 0.05, shown rounded as ± 0.1). The same I is the denominator of each Newton step. It assumes independent games, so the true intervals are somewhat wider (see Limitations).

Q5. How do you know the numbers are right? *(objective)*
`check_fit.py` refits every h from `counts.csv` in pandas/NumPy, independently of the C++ code. All rows of `output.csv` and `openings.csv` match to within 5×10^{-5} rating points.

Q6. Why rating points instead of raw win rates? *(methodological)*
A raw win rate mixes the first move with the rating gap and the draw rate. Rating points correct each game for the gap, so groups with different rating mixes are comparable.

Q7. With millions of games, isn't everything "significant"? *(methodological)*
Yes, so we report effect sizes with 95% CIs and treat differences of a few rating points as small whatever the p-value.

Q8. Why might h rise at high ratings instead of falling? *(interpretive)*
Our guess: weaker players blunder often enough to swamp a small edge, while stronger players convert small edges more reliably. The data show the pattern, not this explanation.

Q9. Why is Correspondence so different ($+2.8 \pm 2.4$)? *(interpretive)*
With days per move, both sides can think through any position, which may blunt the initiative. But it has only 80,774 games and the widest interval, so we do not read much into it.

Q10. Does the opening *cause* the difference in h? *(interpretive)*
Not necessarily. Players choose their openings, and style, preparation and rating differ between groups, so our results show association only.

Section 4: Reflection Essay

This project taught us to turn a vague question ("is White better?") into something measurable: White's advantage as h rating points, fitted by maximum likelihood with confidence intervals, on millions of real games. We learned to stream huge compressed files with a C++ parser, keep the slow counting step separate from the fast analysis, and rebuild every table and figure from one small file.

Unlike textbook problems, the data were messy and nothing told us what the answer should be. We had to decide what to exclude and defend it. With so many games almost every difference was "significant", so the real question became whether an effect was large enough to matter. We also learned that opening choice is not random, so our results show association, not causation.

The open-ended start felt rudderless, but settling on three testable hypotheses (rating, opening, time control) gave us direction. The work was heavier than a normal assignment because data, modelling and writing all had to fit together, and a late change meant re-running everything. It was also the closest we have come to real data work: building a pipeline, checking the numbers, and explaining them to others.

References

Lichess Open Database, `https://database.lichess.org`. Lichess opening names, `https://github.com/lichess-org/chess-openings`. A. E. Elo, *The Rating of Chessplayers, Past and Present*, Arco, 1978. M. E. Glickman, "Example of the Glicko-2 system", 2012.